جامعة خليفة
Khalifa University

Knowledge-and-Language

A Bilingual Recipe Assistant Combining a Knowledge Graph, Lightweight NLP, and LLMs

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Outline

Motivation & Core Challenge

Context:

- Recipe retrieval is often limited to keyword matching.
- Users have complex, multi-criteria constraints (dietary, time, ingredients).
- Multilingual support (PT/EN) is challenging for structured data.

Objective:

Objective

Context:

- Develop a **Bilingual Assistant** that understands intent.
- Combine the **precision of Knowledge Graphs (KG)** with the **fluency of LLMs**.

Objective:

System Pipeline

Hybrid Retrieval: SPARQL (KG) + FAISS (Vector Search)

- **Input:** User query (Portuguese or English).
- **NLP Layer:** Intent classification (**BERT**) + Entity Extraction (**spaCy/RapidFuzz**).
- **Retrieval:** SPARQL for structured data + FAISS index on descriptions.
- **Generation:** LLM (**Groq API**) generates summarized results bilingually.

NLP: Intent & Entity Extraction

1. Intent Classification

- Fine-tuned `bert-base-multilingual-cased`.
- Classes include: `list_by_ingredient`, `list_by_time`, `find_recipe`, etc..
- Achieved ∼92% Intent Accuracy on test queries.

2. Entity Extraction & Linking

- Uses `RapidFuzz` for fuzzy matching against KG labels.

NLP: Intent & Entity Extraction

3. Time Constraint Handling

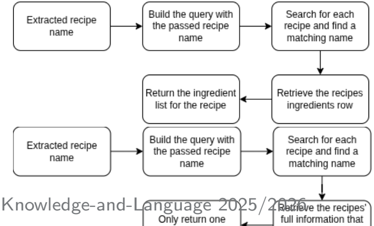
- **Challenge:** User descriptions are highly variable and ambiguous (e.g., "1h20", "under 10 minutes").
- **Solution:** Layered numeric parsing, translation normalization, and regex for exact/maximum duration.
- Time Parsing is robust and handles typos.

SPARQL Functions: Intent-to-Query Mapping

Each user intent is mapped to a specific SPARQL query function, filled with extracted entities, to retrieve structured data from the KG.

Functions for Specific Recipes or Values:

- **Retrieve Ingredients** (`retrieve_ingredients`)
- **Find Recipe** (`find_recipe`)
- **Get Preparation Time** (`get_prep_time`)



Functions for List Retrieval:

- **List by Tag** (`list_by_tag`)
- **List by Ingredient** (`list_by_ingredient`)
- **List by Time** (`query_by_max/exact_minutes`)

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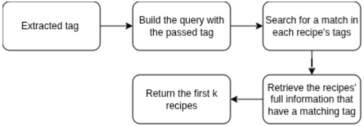

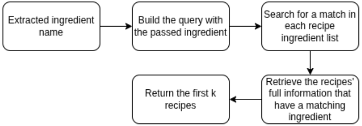

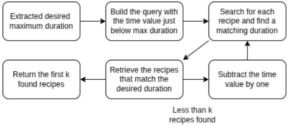

Experimental Results Summary

Key Findings:

- **Bilingual Consistency:** Improved translation normalization ensured stable and deterministic results across PT/EN queries.
- **RAG Impact:** RAG showed little measurable impact; the KG already provided comprehensive, relevant information.
- **Hard Queries:** For creative queries, the LLM used retrieved data to ask clarifying questions, guiding the user back to valid intents.

Conclusion & Future Work

- **Summary:**
  - Successfully integrated a KG with LLMs for a robust bilingual assistant.
  - Solved specific NLP challenges in entity linking, time parsing, and translation consistency.
- **Future Directions:**
  - **Expand KG:** Include more recipes and nutritional facets.
  - **Personalization:** Implement user profiles for dietary restrictions.
  - **Evaluation:** Add automated metrics (BLEU/ROUGE) and reinforcement learning from human feedback (RLHF).

Thank You! Questions?